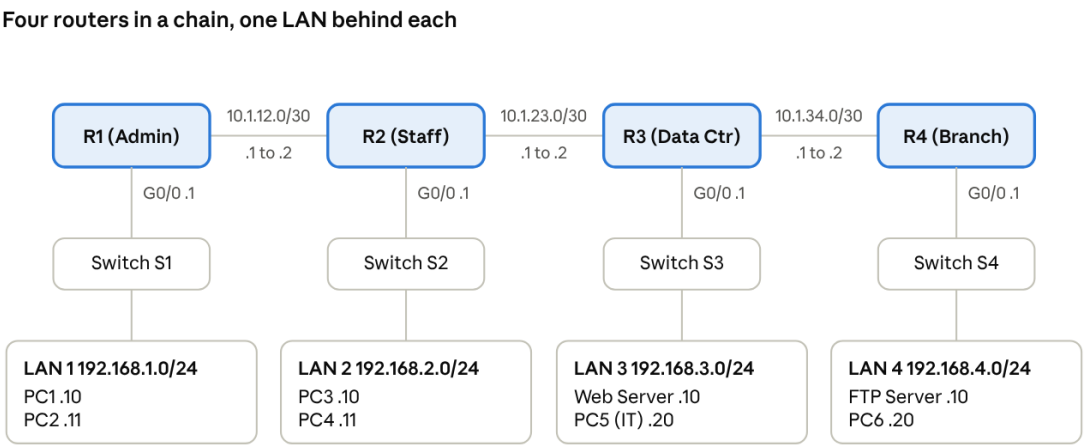
Lab Task: Static Routing and Extended ACLs (Four Routers)

Scenario: A company has four sites. Site 1 (R1) is the Admin office, Site 2 (R2) is the Staff office, Site 3 (R3) is the Data Center with the Web Server, and Site 4 (R4) is a Branch office with the FTP Server. All routing is static. The security team has written four policies (Part D) that must be enforced with extended ACLs.

Tool: Cisco Packet Tracer. **Devices:** 4 x Cisco 2911 routers, 4 x 2960 switches, 6 PCs, 1 Web Server, 1 FTP Server.

Topology and Addressing Table

The four routers are connected in a chain: R1 to R2, R2 to R3, and R3 to R4. Each router has one LAN behind a switch.



lab topology · 4 routers, 4 LANs, 3 point-to-point links

Device	Interface	IP Address	Subnet Mask	Default Gateway	Connected To
R1	G0/0	192.168.1.1	255.255.255.0	N/A	Switch S1 (LAN 1)
R1	G0/1	10.1.12.1	255.255.255.252	N/A	R2 G0/1
R2	G0/0	192.168.2.1	255.255.255.0	N/A	Switch S2 (LAN 2)
R2	G0/1	10.1.12.2	255.255.255.252	N/A	R1 G0/1
R2	G0/2	10.1.23.1	255.255.255.252	N/A	R3 G0/1
R3	G0/0	192.168.3.1	255.255.255.0	N/A	Switch S3 (LAN

Device	Interface	IP Address	Subnet Mask	Default Gateway	Connected To
					3)
R3	G0/1	10.1.23.2	255.255.255.252	N/A	R2 G0/2
R3	G0/2	10.1.34.1	255.255.255.252	N/A	R4 G0/1
R4	G0/0	192.168.4.1	255.255.255.0	N/A	Switch S4 (LAN 4)
R4	G0/1	10.1.34.2	255.255.255.252	N/A	R3 G0/2
PC1 (Admin)	NIC	192.168.1.10	255.255.255.0	192.168.1.1	S1
PC2 (Admin)	NIC	192.168.1.11	255.255.255.0	192.168.1.1	S1
PC3 (Staff)	NIC	192.168.2.10	255.255.255.0	192.168.2.1	S2
PC4 (Staff)	NIC	192.168.2.11	255.255.255.0	192.168.2.1	S2
Web Server	NIC	192.168.3.10	255.255.255.0	192.168.3.1	S3
PC5 (IT)	NIC	192.168.3.20	255.255.255.0	192.168.3.1	S3
FTP Server	NIC	192.168.4.10	255.255.255.0	192.168.4.1	S4
PC6 (Branch)	NIC	192.168.4.20	255.255.255.0	192.168.4.1	S4

The network has seven subnets: four LANs (192.168.1.0/24 to 192.168.4.0/24) and three point-to-point links (10.1.12.0/30, 10.1.23.0/30, 10.1.34.0/30).

Part A: Build the Topology

1. Place four 2911 routers (R1 to R4), four 2960 switches (S1 to S4), six PCs, one Web Server and one FTP Server.
2. Connect R1 G0/1 to R2 G0/1, R2 G0/2 to R3 G0/1, and R3 G0/2 to R4 G0/1 using copper cross-over cables.
3. Connect each router's G0/0 to its LAN switch, and connect the end devices to the switches with copper straight-through cables.
4. Label every device and network address on the workspace.

Part B: Basic Configuration and Services

On all four routers:

1. Set the hostname (R1 to R4) and disable DNS lookup.
2. Set the enable secret to `class` and the console password to `cisco`.
3. Configure lines vty 0 4 with password `cisco`, `login` and `transport input telnet`, so that Telnet can be tested later.

4. Configure the interface IP addresses from the addressing table, add descriptions, and bring the interfaces up.
5. Save the configuration.

On the end devices:

1. Configure IP address, subnet mask and default gateway on all PCs and both servers.
2. On the Web Server, make sure the HTTP and HTTPS services are On. Turn the FTP service Off.
3. On the FTP Server, make sure the FTP service is On (default user `cisco`, password `cisco`). Turn HTTP and HTTPS Off.

Checkpoint B: Every PC can ping its own default gateway.

Part C: Static Routing

No dynamic routing protocol may be used. R1 and R4 are stub routers (only one exit), while R2 and R3 are transit routers in the middle of the chain.

Task C1: R1 (default route). Configure one default static route on R1 pointing to R2.

Task C2: R4 (default route). Configure one default static route on R4 pointing to R3.

Task C3: R2 (next-hop routes). Configure static routes on R2 for every network that is not directly connected: LAN 1, LAN 3, LAN 4 and the link 10.1.34.0/30. Use next-hop IP addresses.

Task C4: R3 (next-hop routes). Configure static routes on R3 for LAN 1, LAN 2, LAN 4 and the link 10.1.12.0/30. Use next-hop IP addresses.

Task C5: Verification.

1. Run `show ip route static` on each router and confirm that every one of the seven subnets is reachable.
2. From PC1, ping PC6 and the FTP Server. From PC6, ping PC1.
3. From PC1, open the web browser and load `http://192.168.3.10`.
4. From PC1, run `tracert 192.168.4.10` and record the hops.

Checkpoint C: Full end-to-end connectivity must work before you configure any ACL. Save screenshots of the routing tables.

Part D: Extended ACL Rules

Use extended ACLs only (numbers 100 to 199, or named extended ACLs). An extended ACL can match source, destination, protocol and port, so it is placed as close to the source as possible. Remember the implicit `deny any` at the end of every ACL. Only one ACL is allowed per interface, per direction.

Rule	Policy	ACL Type and ID	Apply On	Direction
1	LAN 1 may reach the Web Server (192.168.3.10) only through HTTP (TCP 80) and HTTPS (TCP 443). Any other traffic from LAN 1 to the Web Server, including ping, is	Numbered, 110	R1 G0/0	In

Rule	Policy	ACL Type and ID	Apply On	Direction
	denied. All other LAN 1 traffic is allowed.			
2	PC3 (192.168.2.10) must not use FTP (TCP 21) on the FTP Server (192.168.4.10), but may still ping it. No host in LAN 2 may Telnet (TCP 23) to any device. All other LAN 2 traffic is allowed.	Named, STAFF_POLICY	R2 G0/0	In
3	The Web Server must not start any connection to LAN 1 or LAN 2. It may only send replies from its web ports (source TCP 80 and 443). All other LAN 3 traffic is allowed.	Named, PROTECT_WEB	R3 G0/0	In
4	LAN 4 hosts must not send ping requests (ICMP echo) to LAN 1. LAN 1 must still be able to ping LAN 4, so echo replies must pass. All other LAN 4 traffic is allowed.	Numbered, 140	R4 G0/0	In

Task D1. Write ACL 110 for Rule 1. The two permit statements for the web ports must come before the deny statement for the Web Server.

Task D2. Write the named ACL STAFF_POLICY for Rule 2. Both parts of the policy go into the same ACL, because only one inbound ACL can sit on R2 G0/0.

Task D3. Write the named ACL PROTECT_WEB for Rule 3. This rule matches on the source port of the server (`eq 80` placed after the source address), not the destination port.

Task D4. Write ACL 140 for Rule 4. Use the ICMP message type `echo`, not the whole ICMP protocol.

Task D5. Run `show access-lists` on each router and `show ip interface G0/0` to confirm placement and direction. After Part E, record the match counters next to each statement.